

Tanvir Ahmed

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RESEARCH INTERESTS

Biomedical Image Analysis • Quantum Machine Learning • Trustworthy Deep Learning • Computer Vision

RESEARCH SUMMARY

Computer science graduate and independent researcher in machine learning and computer vision. My main line of work is deep learning for medical images, covering colorectal histopathology, bone-marrow cytology, tuberculosis microscopy and two-photon calcium imaging. A second line is quantum machine learning: feature selection written as a QUBO and solved with QAOA, variational circuits used as classifiers, and training-free quantum kernels, run on simulators and on IBM Quantum hardware. In both areas I spend most of my effort on evaluation that holds up, which in practice means patient-disjoint splits, leakage audits, calibration and per-class results rather than a single headline score. Code is public for every project.

EDUCATION

Independent University, Bangladesh

January 2015 — December 2019

Department of Computer Science and Engineering

Bachelor of Science in Computer Science, Minor in Environmental Management

CGPA: 2.92 out of 4.00 (Final-year GPA: 3.07)

PUBLICATIONS

Journal paper

Under Review — Major Revision Submitted

- Abhijit Kumar Ghosh, Ar Rafiul Islam Rafim, Fahmid Al Farid, **Tanvir Ahmed**, Enam Ahmed Taufik, Jia Uddin, and Hezrul Abdul Karim. “Gated Dual-Scale Feature Fusion for Colorectal Cancer Grading in Centralized and Federated Pathology.” *Informatics in Medicine Unlocked*, 2026. Major revision submitted. Code: [GitHub].

Conference paper

Accepted

- Dhiman Kumar Sarker, Fyaz Nafin Rahman, Md Rezaul Karim Khan, Sushmoy Paul Rohit, **Tanvir Ahmed**, and Md Mizanur Rahman. “Real-Time Browser Threat Detection Security Agent via Local VLM-LLM.” Accepted at the *5th IEEE International Conference on Robotics, Automation, Artificial Intelligence and Internet-of-Things* (RAAICON 2026), 2026.

Submitted

- Tanvir Ahmed**, Jannatul Ferdous Shorna, Md Mahbubur Akash, Md Khorshed Alam, and Md Tarek Hasan. “DARE-FS: A Selector-Agnostic Operator for Detecting and Repairing Class-Specific Feature-Selection Harm.” Manuscript submitted to the *38th IEEE International Conference on Tools with Artificial Intelligence* (ICTAI 2026), 2026. Paper ID: 778.
- Swagotam Malakar, **Tanvir Ahmed**, Abhijit Kumar Ghosh, Mohammad Zavid Parvez, Prabal Barua, and Subrata Chakraborty. “Beyond Confidence: A Leakage-Audited Selective-Prediction Study of Acid-Fast Bacilli Detection in Ziehl-Neelsen Microscopy.” Manuscript submitted to the *International Conference on Digital Image Computing: Techniques and Applications* (DICTA 2026), 2026. Paper ID: 79.
- Md Faisal Sheikh, **Tanvir Ahmed**, Enam Ahmed Taufik, Abhijit Kumar Ghosh, Mohammad Zavid Parvez, Prabal Barua, and Subrata Chakraborty. “Towards Trustworthy Tuberculosis Detection: Subgroup Calibration of Object Detectors in Sputum-Smear Microscopy.” Manuscript submitted to the *International Conference on Digital Image Computing: Techniques and Applications* (DICTA 2026), 2026. Paper ID: 80.

PROJECTS

PCMMMD: Federated Learning and Knowledge Distillation for Plasma Cell Detection [GitHub]

- Benchmarked centralized, few-shot, knowledge-distillation (EfficientNet-B0 teacher to MobileNetV3 student), and federated (FedAvg/FedProx, one client per patient) regimes for plasma-cell classification of bone-marrow crops in multiple myeloma, under patient-disjoint folds, validation-only model selection, and image-hash de-duplication to eliminate leakage.

Direct-WAM: Self-Supervised Denoising for Two-Photon Calcium Imaging [GitHub]

- Proved a bias-floor theory of self-supervised objectives (per-objective error floor, ensemble bias-crossing, distillation transfer bound) and produced a corrected AI4Life CIDC25 benchmark in which a faithfully retrained DeepCAD-RT reaches +23.65 dB stSNR, while the F_0 -free Direct-WAM 3D U-Net distilled from a 13-way ensemble reaches +18.15 dB.

QCIDS: Quantum-Classical Feature Selection for Network Intrusion Detection [GitHub]

- Cast intrusion-detection feature selection as a QUBO solved by QAOA ($p = 1, 2, 3$) on simulators and IBM 156-qubit hardware, simulated annealing, and classical baselines under matched leakage-free cross-validation; at $k = 15$ on NF-ToN-IoT, QAOA $p = 2$ held macro-F1 at 0.763 where greedy mutual information degraded from 0.760 to 0.718.

Q-PlasmaNet: Hybrid Classical-Quantum Classifier for Plasma Cell Recognition [GitHub]

- Paired a frozen CNN backbone with a 4–8 qubit data re-uploading variational quantum head (PennyLane) at matched parameter budget, reaching F1 0.927 with 178 trainable parameters — statistically tied with a parameter-matched MLP (0.932) and below a 2.2M-parameter MobileNetV2 (0.949), establishing parity rather than quantum advantage.

CovNCG: Training-Free Dense Quantum Feature Maps [GitHub]

- Designed a training-free quantum kernel from non-commuting multi-feature generator blocks (three features per two qubits) with a closed-form kernel-variance result scaling as c^4 , turning avoidance of exponential concentration into a bandwidth tuning rule; evaluated with a projected kernel and SVM on five medical datasets in Qiskit with IBM QPU validation.

RESEARCH EXPERIENCE

Independent Researcher

Independent and collaborative research

Boca Raton, Florida, USA

January 2025 — Present

- Author and co-author five manuscripts with international collaborators, one accepted and four under review, across medical imaging, quantum machine learning and feature selection.
- Run the full experimental pipeline on each project: auditing the data for leakage, building and training the models, benchmarking against classical and quantum baselines, and reporting results over multiple seeds with confidence intervals.

PROFESSIONAL EXPERIENCE

Upwork & Fiverr

Freelance Web Developer and E-commerce Specialist

Remote — Dhaka, Bangladesh

December 2014 — November 2022

- Delivered e-commerce storefronts for international clients, both as customized Shopify builds (themes, product catalogues, checkout flows) and as custom-developed websites written from scratch to client specification.
- Ran day-to-day store operations across order and inventory management, customer inquiries, and issue resolution, sustaining repeat client relationships over an eight-year freelance practice.

IDOL Focus — A Professional Training Institute

Web Developer Intern

Dhaka, Bangladesh

September 2019 — December 2019

Supervisors: Md. Asif Bin Khaled, Lecturer, Department of Computer Science and Engineering, IUB (academic); Md. Al Mamun, IDOL Focus (industry)

- Built the Admin and Agent modules of an online job portal (*idolfocusjobs.com*) serving four user roles: role-based authentication, jobseeker and employer account management, CV database administration, job-posting approval, and payment handling.
- Implemented in PHP, JavaScript, Ajax, jQuery, and Bootstrap over MySQL; documented in an undergraduate internship report submitted toward the BSc degree.

TECHNICAL SKILLS

Programming:	Python, SQL, PHP, JavaScript
ML Frameworks:	PyTorch, Scikit-learn, LightGBM, TensorFlow, OpenCV, NumPy, Pandas, Matplotlib
Quantum ML:	PennyLane, Qiskit, QAOA/QUBO formulation, variational circuits, quantum kernels, IBM Quantum hardware
Core Areas:	Machine Learning, Computer Vision, Deep Learning, Federated Learning, Knowledge Distillation
Databases:	MySQL, phpMyAdmin
Web / Software:	HTML, CSS, JavaScript, jQuery, Ajax, Bootstrap, Shopify, Apache Web Server
Tools:	L ^A T _E X, Git, Jupyter Notebook, Google Colab
Languages:	Bangla (native), English (fluent)